MILITARY SPECIFICATION

INK, REACTANT

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for one type of reactant ink.

2. APPLICABLE DOCUMENTS.

2.1 Government documents

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

	BB-N-411	Nitrogen, Technical
*	DD-B-585	Bottles and Jars, Glass, Prescription
*	MMM-A-180	Adhesives, Polyvinyl Acetate Emulsion

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 5321), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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SPECIFICATIONS

Military

MIL-E-52171	Ethylene Oxide, Technical

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

* (Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

MUNSELL COLOR COMPANY

2.5R Chart	Glossy Book of Color

(Application for copies of color chips should be addressed to Munsell Color Company, Inc., 2441 North Calvert Street, Baltimore, MD 21218.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 3951	Packaging, Commercial

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* (Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3 1 First article When specified (see 6 2), a sample shall be subjected to first article inspection (see 6 3) in accordance with 4.3.

3.2 Preparation The ink shall be prepared as specified in 3.2.1 through 3.2 2.

3 2 1 Ingredients

3.2 1 1 Reagent grade alcohol. The alcohol mixture shall consist of 95 ± 2 percent (by volume) of ethyl alcohol denatured with 5 ± 1 percent (by volume) of isopropyl alcohol The residue after evaporation shall be less than 0.001 percent by weight

3 2.1 2 Benzene Analytical reagent grade shall not have impurities exceeding the limits specified in Table I.

TABLE I Maximum Impurities in Analytical Grade Benzene

Impurity	Percent (by weight)
Water	0.0200
Sulfur compounds	0 0050
Thiophene	0 0005
Residue after evaporation	0 0010

3.2 1 3 Indicator. The indicator shall be 1,8 Dihydroxyanthraquinone (Chrysazin) having a melting point between 192°C and 194°C (378°F to 381°F)

3.2.1 4 Magnesium bromide hexahydrate The chemical shall be crystal purified grade or better, with the compound consisting of small, colorless, odorless, crystals not exceeding 1/8 inch in two dimensions (length excluded). Do not use if a HBr, Br_2, or Cl_2 odor is detected. Spread the amount used for a given ink lot on a clean glass pan and dry in a circulating oven at 70 ± 5°C (158 ± 9°F) for 3 ± 1 hours prior to using.

* 3 2 1.5 Adhesive The adhesive shall conform to the requirements of MMM-A-180, Class B.

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3.2.2 Procedure.

(a) The atmosphere in which the ink shall be prepared shall be acid and alkaline free (pH of 6 ± 1).

(b) Dissolve 50 ± 1 grams $MgBr_2.6H_2O$ in 100 ± 2 milliliters of reagent alcohol.

(c) Dissolve 75 ± 1 milligrams (mg) of 1,8-Dihydroxy-anthraquine (chrysazin) in 20 ± 1 milliliters benzene

(d) Pour solution from (c) into solution from (b) and shake until mixed and measure resultant volume to the nearest milliliter.

(e) To each 115 ± 1 milliliters of the solution in (d) add 69 ± 1 milliliters adhesive. Shake thoroughly until a homogenous yellow emulsion is obtained. Store in a clean airtight amber glass bottle (conforming to DD-B-585, Type II, Style 1, Color Amber, except that cover liner shall be aluminum faced).

CAUTION: Filled bottles shall be kept tightly closed at all times except when ink is being removed. Also, do not use any method involving heat during preparation.

NOTE: Ink may be made in multiples of amounts specified in 3 2 However, the given tolerance of each measurement remains the same. All mixing and reagents shall be at 22.2 ± 1.1°C (72 ± 2°F).

3.3 Performance.

3.3.1 Color fastness. When tested in accordance with 4.6.1 the color of the sample shall be as yellow as, and no more pink than color 23613 of FED-STD-595.

3.3.2 Color change. When tested in accordance with 4.6.2 the color of the sample shall change from an opaque yellow to a standard red 7/6 to 6/8 as described in Munsell Color Standard, Chart 2.5R.

3.4 Storage life. The ink shall be stored in accordance with 3.2.2(e) and shall be used within 90 days of preparation.

3.5 Workmanship. The ink, when examined visually, shall be a homogenous liquid, free of lumps, of any separated solids, or evidence of phase separation.

4. QUALITY ASSURANCE PROVISIONS

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4 1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

* 4 1 1 Responsibility for compliance All items must meet all requirements of Sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material

4 2 Classification of inspection The inspection requirements specified herein are classified as follows.

a. First article inspection (see 4 3)
b Quality conformance inspection (see 4.4).

* 4 3 First article inspection. The first article inspection of the ink shall consist of a visual inspection (4.5 1) and a performance test (4.5.2).

* 4.3.1 First article samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, the contractor shall submit three test samples prepared in accordance with 4.3.1.1 from each of five 250 milliliter bottles of ink prepared in accordance with 3.2 The samples shall be representative of the components, materials and workmanship to be used during production When a contractor is in continuous production of the ink from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2 1). Approval of the first article inspection samples or the waiving of the first article inspection does not waive the requirements of performing the quality conformance inspection The first article inspection samples shall be furnished to the Government as directed by the contracting officer (6.2 1) The first article inspection sample shall be identified by securely attached tags or labels durably marked with the following information:

Sample for First Article Inspection
INK, REACTANT
Name of manufacturer
Name of distributor (if applicable)
Product designation (manufacturer's)
Date of manufacture
Submitted by (name) (date) for first article
inspection in accordance with the requirements
of MIL-I-81971A(AS).

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4.3.1.1 Preparation of test samples. The test samples shall be prepared as follows in an atmosphere having a pH of 6 ± 1:

(a) Filter paper for the test samples shall be either Whatman number 42, Schleicher and Schuell number 589, Munktill number OOH, or equivalent approved by the procuring activity.

(b) The paper shall be cut into disks 1.312 +0.000 inch and -0.060 inch diameter.

(c) Apply a uniform coating of 55 ± 20 mg to each sample disk. Coating weight shall be determined after the sample has been dried in 6 percent relative humidity for a minimum of 24 hours.

(d) Place the coated disks in sealed clear gas bottles and hold for not less than 7 days.

CAUTION: Materials (such as gaskets) used to seal bottles shall be free of any basic or acid outgassing properties.

(e) Examine the disks for color in accordance with 3.3 1.

* 4.3.2 Test report. Two copies of the contractor's test report, containing complete test data for the end item and referring specifically to the applicable paragraphs, shall be submitted with the first article inspection sample.

4.4 Quality conformance inspection. The quality conformance inspection shall be performed on five test samples from each lot of ink prepared and tested in accordance with 4.3.1.1 and tested in accordance with 4.5 through 4.6. A lot is defined as a lot of ink prepared in accordance with 3 2.2 from one lot of each ingredient as defined in MIL-STD-105. A lot shall be unacceptable if a sample fails to meet any one of the test requirements specified.

* 4.4.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified

4.5 Inspection methods.

4.5.1 Visual inspection. The test samples prepared in accordance with 4.3.1.1 shall be placed in closed clear bottles (see 3 2.2(e)) and stored for 7 days at 22.1 ± 1.1°C (72 ± 2°F). The atmosphere in the bottles shall be the atmosphere that was in the test sample preparation area at the time of the sample preparation.

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4.5.2 Performance test. The test samples prepared in accordance with 4.3.1 1 shall be tested as follows:

(a) Verify that all test equipment is calibrated before beginning the test. Testing should be done in an environment having a pH of 6 ± 1.

(b) Place samples in a room with 6 percent maximum relative humidity

(c) Clean test vessel by immersion for a minimum of 5 minutes in hot tap water and then in deionized or distilled water Rinse the inner surfaces successively with analytical reagent grade dioxane, chloroform, and ethanol. Dry in an oven at 110 ± 5°C (230 ± 9°F) for 1/2 hour minimum and cool to room temperature.

NOTE Each test vessel shall be such that the ratio of the volume of the test vessel (in liters) to the total surface area of the samples (in square inches) shall be a minimum of 1 5. The test vessel shall have provisions for purging, for inserting the samples, for injection of measured amounts of ethylene oxide, and for viewing the test samples as required for detection of color change

(d) Purge vessel for a minimum of 5 minutes in dry nitrogen conforming to BB-N-411 (less than 1 percent relative humidity) Place the open test vessel in the 6 percent maximum relative humidity atmosphere for a minimum of 12 hours

(e) Insure that the samples have been in the relative humidity atmosphere for a minimum of 12 hours.

(f) Place the samples in the empty test vessel and slide cover into position

(g) Add distilled water as necessary to obtain relative humidity between 14 and 18 percent

(h) Allow vessel with test samples to stabilize for at least 24 hours Test relative humidity of atmosphere inside test vessel and adjust as necessary to obtain relative humidity between 14 and 18 percent

(i) Purge the fuel line with ethylene oxide conforming to MIL-E-52171 and connect to test vessel

(j) Draw a partial vacuum to 3 percent of ambient atmospheric pressure (in mm of mercury) ± 0 3mm on the test vessel. Inject sufficient ethylene oxide to obtain a 2 3 ± 0.1 percent concentration (by volume) The injected stream of gas shall not strike the test samples.

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(k) Analyze the test atmosphere for ethylene oxide concentration between 1 and 2 hours later. The concentration shall be at least 2 percent.

(l) The samples shall change color from yellow to red (see 3.2.2) within 24 hours.

4.6 Preservation, Packing and Marking. The preservation, packing and marking shall be in accordance with Section 5. The inspection level shall be S-2 and the Acceptable Quality Level (AQL) of 4.0

5. PACKAGING.

5.1 Preservation. Preservation shall be commercial.

5.1.1 Commercial The ink shall be preserved in accordance with ASTM D 3951

5.2 Packing. Packing shall be commercial.

5.2.1 Commercial. The bottles of ink shall be packed in such a manner, that they are protected against breakage or damage during direct shipment from the contractor to the first receiving activity. The ink shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with applicable requirements of MIL-STD-129.

5.3.1 Precautionary markings. Each bottle of ink shall be marked with the following warnings:

1. Lot number and date of manufacture.

2. Use ink within 3 months of date of manufacture

3. Filled bottles shall be kept tightly closed at all times except when ink is being removed.

4. Do not use any method involving heat during preparation.

6. NOTES

6.1 Intended use. This reactant ink is intended for use as a coating on a leak indicator disk. The coated disk is used to detect ethylene oxide vapor

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) First Article sample size if other than specified in 4.3.1.

(c) Responsibility for the performance of the First Article Inspection.

(d) Selection of applicable levels of preservation and packing

(e) Marking if other than specified.

* 6 3 First article. When a first article inspection is required, the ink should be a first article sample The first article sample should consist of the sample specified in 4.3 1 The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

* 6 4 Subject term (keyword) listing

Ethylene oxide
First article inspection
Ink
Reactant

6.5 Changes from previous issue The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing Activity
Navy - AS
(Project No 6630-N467)

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